DIELECTRIC

Insulating materials which may not conduct electricity through it but induced charges are produced on its faces in the presence of an external electric field. Such types of insulator are called Dielectric.

Dielectric is an insulating material, which transmits electric effects without actual flow of electric charge carriers. Examples of such dielectric material are waxed paper, glass, mica, different plastics etc.

Dielectric Constant: The dielectric constant is the property of the dielectric material and varies from one material to another and denotes a large-scale property of dielectrics without specifying the electrical behaviour on the atomic scale. *Dielectric constant of dielectric medium is the ratio of the applied electric field (E_o) to that of reduced value of electric field (E) inside the dielectric material.*

$$k = \frac{\vec{E}_o}{\vec{E}} \qquad\qquad \vec{E} < \vec{E}_o \Rightarrow k > 1$$

Dielectric constant is a number without any dimensions and the value of the static dielectric constant of any material is always greater than one. Its value is one for a vacuum or air.

In terms of capacitance, Dielectric constant of dielectric material is equal to the ratio of the capacitance (C) of a capacitor filled with the dielectric material to the capacitance (Co) of an identical capacitor without the dielectric material.

$$k = \frac{C}{C_o}$$

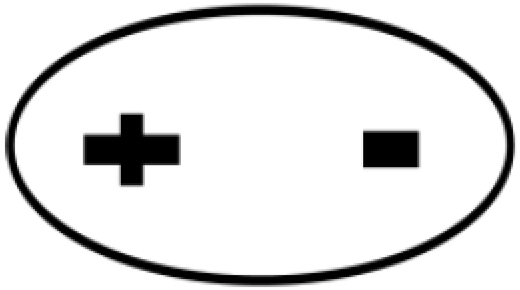
In terms of permittivity, Dielectric constant is the ratio of the permittivity (ε) of a material to the permittivity (ε_o) of free space.

$$k = \frac{\varepsilon}{\varepsilon_o}$$

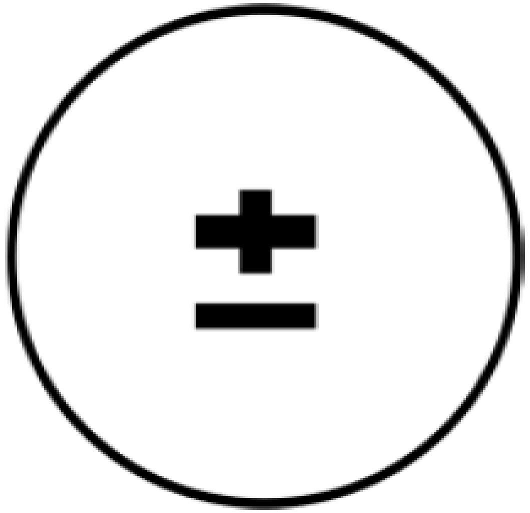
CLASSIFICATION OF DIELECTRIC ON THE BASIS OF MOLECULAR STRUCTURE

On the basis of the molecular structure, dielectric material is classified in two types:

1) Polar Dielectrics
2) Non polar Dielectrics



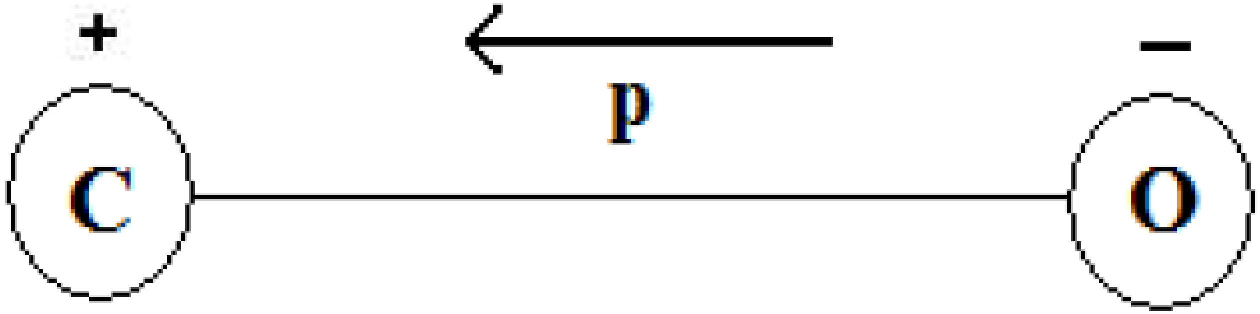
Polar Dielectrics



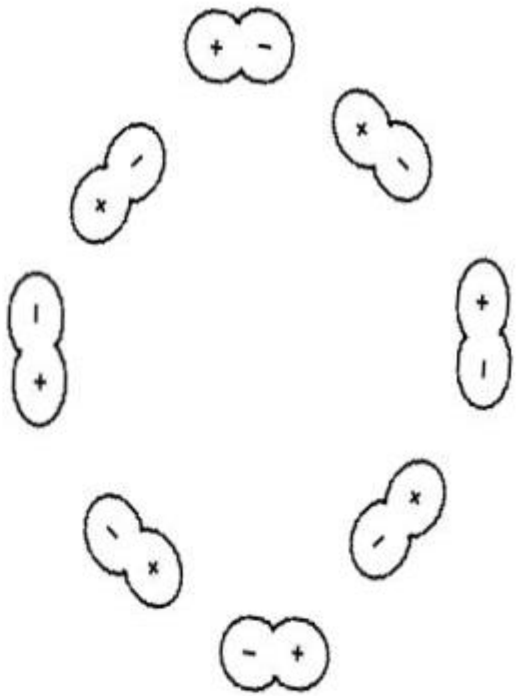
Non polar Dielectrics

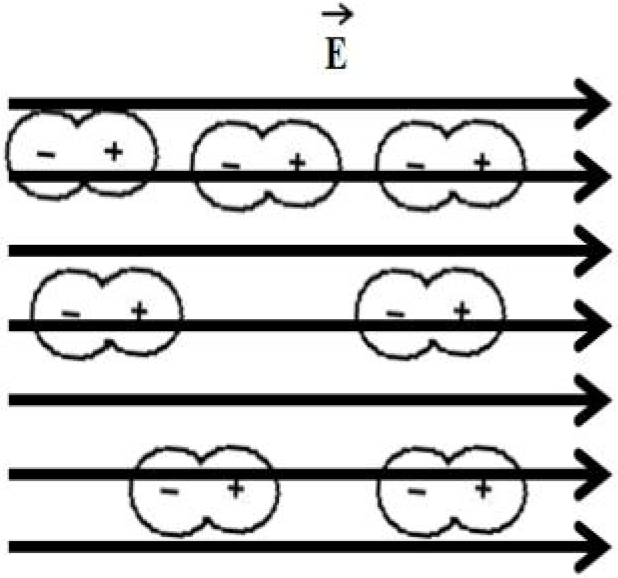
Polar dielectrics

Polar dielectric are those in which centre of gravity of positive and negative charges do not coincides with each other and having permanent electric dipole moment



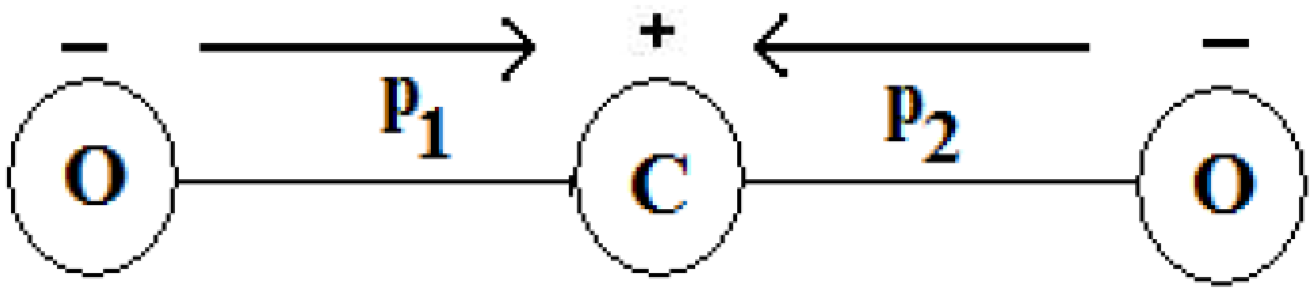
Effect of electric field on Polar molecule



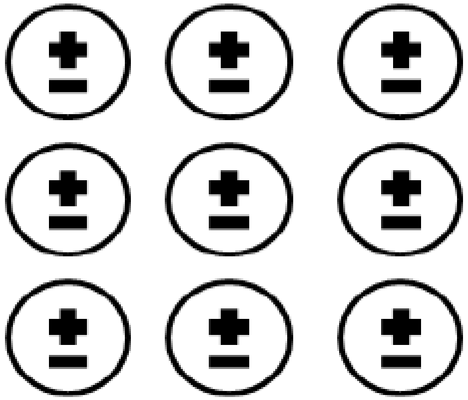


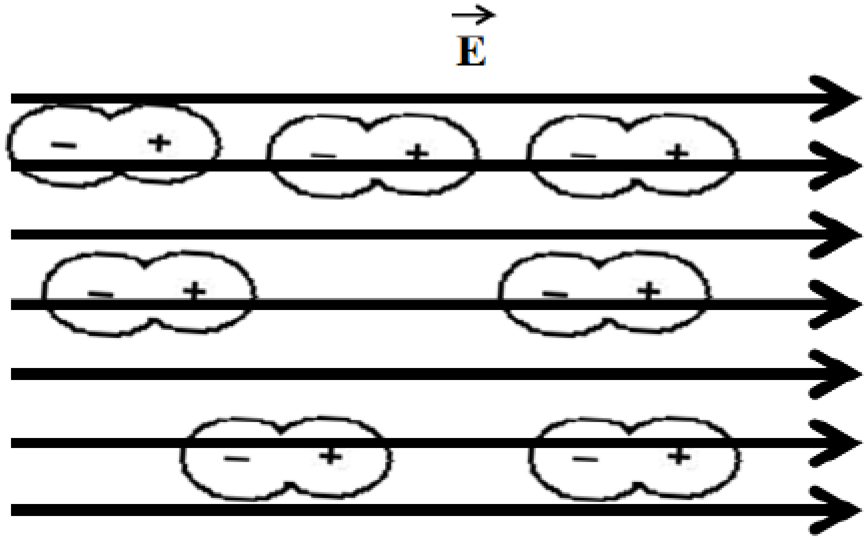
Non Polar dielectrics

Non-Polar dielectrics are those in which centre of gravity of positive and negative charge coincides with each other and having zero electric dipole moment.



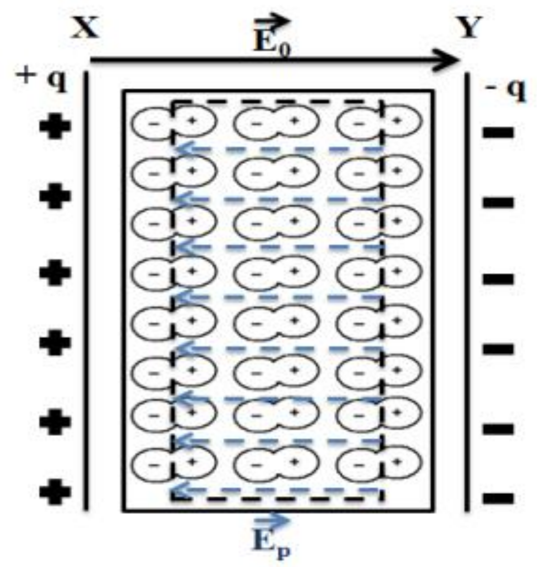
Effect of electric field on Non-Polar molecule

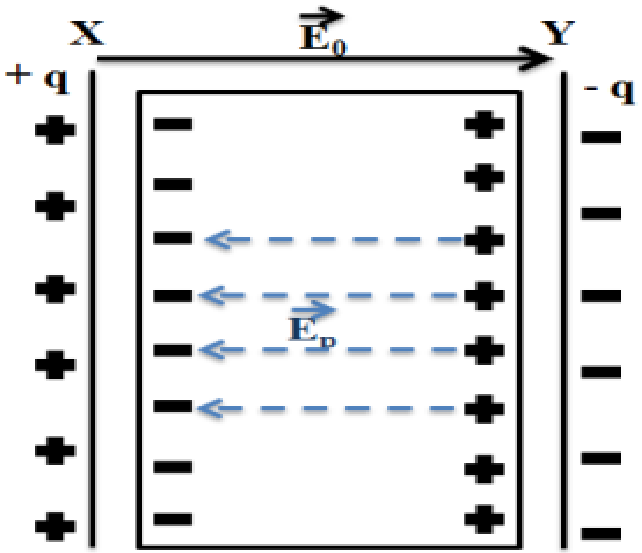




POLARISATION OF DIELECTRIC

The polarisation is the stretching of dielectric atoms or molecule due to the displacement of charges in the atoms under the influence of applied electric field.





The phenomenon of appearance of the charge on the surface of dielectric slab, induced by an external electric field is defined as polarisation.

Net electric field inside the dielectric will be $\vec{E} = \vec{E}_o - \vec{E}_p$

ELECTRIC POLARISATION VECTOR

Electric Polarisation vector measure the extent to which the molecules of a dielectric becomes polarised by an electric field, or oriented in the direction of the electric field. Mathematically it is the ratio of dipole moment per unit volume.

$$\vec{P} = \frac{\vec{p}}{V}$$

If d is the separation between the induced charges or the thickness of dielectric slab and q is the magnitude of induced charge on the surface of dielectric slab; then induced dipole moment $\vec{p} = q.\vec{d}$.

Equation in scalar form can be written is

$$P = \frac{p}{V} = \frac{q.d}{a.d} = \frac{q}{a} = \sigma_p$$

$$P = \sigma_p$$

Electric susceptibility χ_e

Electric susceptibility χ_e of a dielectric material is a measure of how easily it polarizes in the presence of an electric field. It is an important physical quantity to describe the behavior of a dielectric in electric field. Experimentally, it is found that polarization is directly proportional to the net electric field inside the dielectric, i.e.

$$\vec{P} \,\alpha\, \vec{E}$$

or $\vec{P} = \varepsilon_o \chi_e \vec{E}$ or in scalar form $P = \varepsilon_o \chi_e E$

$$\chi_e = \frac{P}{\varepsilon_o E}$$

The value of χ_e is zero for vacuum as there is no polarisation in vacuum.

Units of Electric Susceptibility: No S.I. unit.

ATOMIC POLARIZABILITY

Atomic polarizability α *is defined as the induced dipole moment per unit electric field.*

$$\boxed{\alpha = \frac{p}{E_o}}$$

EQUATION OF CONTINUITY

Consider a closed surface A enclosing a volume V in which current I is flowing (Figure 6.4). Consider a small surface $d\vec{a}$ on the surface A. If $\vec{J}$ is the current density, then total current flowing through the closed surface A

$$I = \oiint_A \vec{J}.d\vec{a} \qquad (6.5.1)$$

Let ρ be the volume charge density, then total charge in the volume V is given by

$$q = \iiint_V \rho \, dV \qquad (6.5.2)$$

Since the current is flowing outward, it means that charge within closed surface is decreasing with time. The time rate of decrease of charge i.e., current within volume V is

$$I = -\frac{d}{dt}\iiint_V \rho \, dV \qquad (6.5.3)$$

Hence Equation (6.5.1) can be written as

$$\oiint_A \vec{J}.d\vec{a} = -\frac{d}{dt}\iiint_V \rho \, dV$$

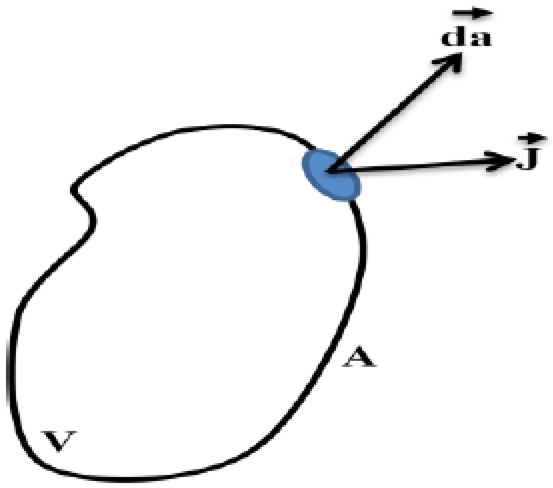


Figure 6.4

or $$\oint\!\!\!\oint_A \vec{J}.d\vec{a} = -\iiint_V \frac{d\rho}{dt} dV \qquad (6.5.4)$$

According to divergence theorem

$$\oint\!\!\!\oint_A \vec{J}.d\vec{a} = \iiint_V \vec{\nabla}.\vec{J}\, dV \qquad (6.5.5)$$

Compare Equation (6.5.4) and (6.5.5)

$$\iiint_V \vec{\nabla}.\vec{J}\, dV = -\iiint_V \frac{d\rho}{dt} dV$$

$$\vec{\nabla}.\vec{J} = -\frac{d\rho}{dt}$$

$$\vec{\nabla}.\vec{J} + \frac{d\rho}{dt} = 0 \qquad (6.5.6)$$

Equation (6.5.6) is called the Equation of continuity and holds at each point of space and time. It is also called the mathematical form of law of conservation of charge.

Special case

If the current is steady, i.e. magnitude of charge is not changing with space or time. Or total amount of charge entering the volume V is equal to total amount of charge leaving the volume V. Thus, there is no change in the volume charge density with time, hence

$$\frac{d\rho}{dt} = 0$$

Hence Equation (6.5.6) become

$$\vec{\nabla}.\vec{J} = 0$$

Thus for steady current, divergence of current density is zero.

UPES

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t} \qquad \text{(ii)}$$

This is called the continuity of current equation.

Effect of introducing charge at some interior point of a conductor/dielectric

According to Ohm's law

$$\mathbf{J} = \sigma \mathbf{E}$$

According to Gauss's law

$$\nabla \cdot \mathbf{E} = \frac{\rho_v}{\varepsilon}$$

Equation (ii) now becomes

$$\nabla \cdot \sigma \mathbf{E} = \frac{\sigma \rho_v}{\varepsilon} = -\frac{\partial \rho_v}{\partial t}$$

or $$\frac{\partial \rho_v}{\partial t} + \frac{\sigma}{\varepsilon}\rho_v = 0$$

This is homogeneous liner ordinary differential equation. By separating variables we get

$$\frac{\partial \rho_v}{\rho_v} = -\frac{\sigma}{\varepsilon}\partial t$$

Integrating both sides

$$\ln \rho_v = -\frac{\sigma t}{\varepsilon} + \ln \rho_{vo}$$

where $\ln \rho_{vo}$ is a constant of integration

$$\rho_v = \rho_{vo} e^{-t/T_r} \qquad \text{(iii)}$$

where

$$T_r = \frac{\varepsilon}{\sigma}$$

ρ_{vo} is the initial charge density (i.e., ρ_v at $t = 0$)

Equation (iii) shows that as a result of introducing charge at some interior point of the material there is a decay of the volume charge density ρ_v.

The time constant T_r is known as the relaxation time.

Relaxation time is the time in which a charge placed in the interior of a material drops to e^{-1} = 36.8 % of its initial value.

For Copper $T_r = 1.53 \times 10^{-19}$ sec (short for good conductors)
For fused Quartz T_r = 51.2 days (large for good dielectrics)